

FINAL PROJECT REPORT

Radial Weather & Happiness Change

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Final Term Project Report

1. Introduction

This project investigates how **weather conditions relate to changes in my happiness before and after playing golf**. Over the 12 weeks of the term (September 14 – December 7, 2025), I recorded data for **23 golf rounds**, including:

- Date
- Golf course
- Weather description
- Happiness Before (1–10)
- Happiness After (1–10)

The final visualization uses a **radial layout**, where each golf round is drawn as a “spoke.” This format highlights the emotional change from before to after the round and reveals how weather correlates with that change.

The purpose of this report is to document the data collection process, rationale for visualization design, alternative sketches, technical implementation, insights gained, and future directions for data collection.

2. Data Collected

2.1 What Data I Collected

For every golf round, I recorded:

- **Happiness Before** playing (1–10)
- **Happiness After** playing (1–10)
- **Weather condition** (as observed at the course)
- **Golf course location**
- **Date**

The **happiness scale** was self-defined:

- **1 = very unhappy**
- **10 = extremely happy**

Weather descriptions (e.g., “Light Rain,” “Partly Sunny”) were later **categorized into three groups** for visualization:

- **Sunny / Partly Sunny**
- **Cloudy / Overcast**
- **Rainy / Drizzle / Light Rain**

2.2 Why I Collected This Data

Golf is an important personal activity for me and often affects my emotional state. I wanted to explore:

- Does weather influence mood improvement from golf?
- Are sunny days really “happier days”?
- Do rainy days reduce enjoyment or emotional benefit?

The goal was **not to analyze golf performance**, but to focus purely on **mood change**.

2.3 Frequency

I collected one entry **per golf round**. The total dataset includes **23 rounds**, covering the entire course duration up to Week 12.

The full dataset appears in **Appendix A**.

3. Question & Hypothesis

Research Question

Does weather condition relate to the difference between my happiness before and after each golf round?

Hypothesis

Sunny or partly sunny weather will correspond to **larger positive increases** in happiness, while cloudy or rainy days will correspond to **smaller gains**.

4. Final Visualization Overview

Visualization Title

Radial Weather & Happiness Change

Description

The final visualization is a **radial spoke chart**, where each round is positioned at an angle around the center. Each spoke displays:

- **Red segment:** Happiness Before
- **Green segment:** Happiness After
- **Colored circle at tip:** Weather category
- **Radial distance:** Happiness value (1–10)

Encoding Choices

Visual Element	Data Represented
Angle	Round identity
Red line	Happiness Before
Green line	Happiness After
Circle color	Weather (Sunny / Cloudy / Rainy)
Circle position	Happiness After value
Distance from center	Numerical happiness score

Why Radial Design Fits the Project

The radial layout reflects a **golf-themed structure** (like looking at the sun or a golf ball's energy lines). It also:

- Emphasizes **difference** between before and after
- Shows many rounds without visual clutter
- Supports **multi-variable encoding** in one figure
- Places all rounds around the same origin → easy comparison

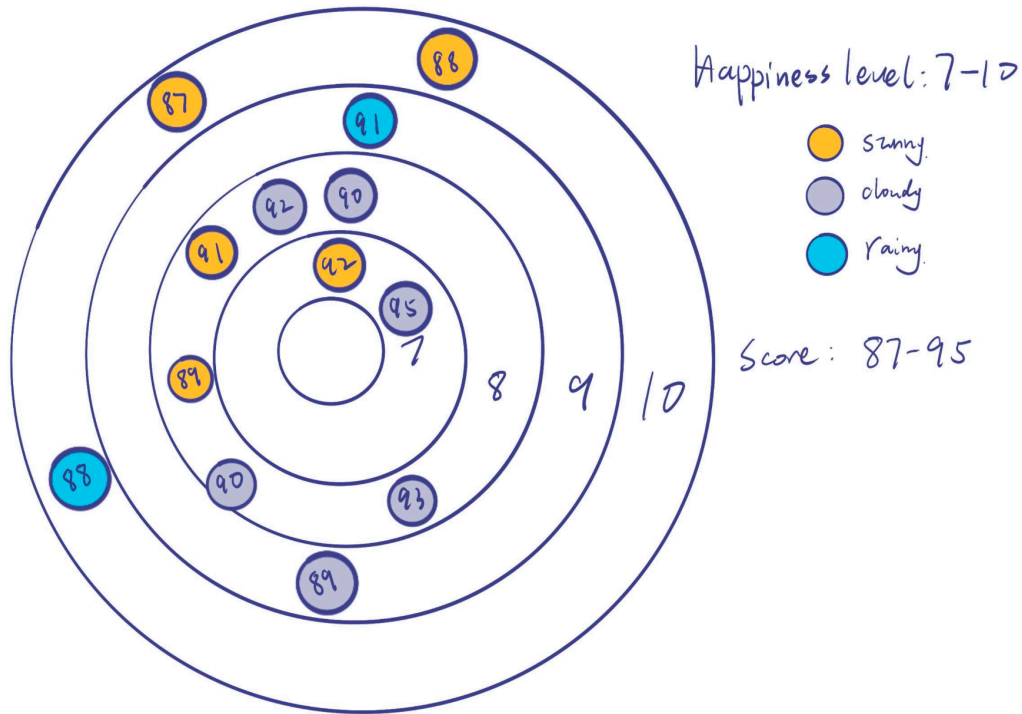
Unlike bar charts or timelines, the radial form makes emotional change visually intuitive.

5. Alternative Visualization

The assignment requires at least one alternative visualization that *conceptually works* but was *not used*.

Alternative Sketch Used (Sketch 2)

Sketch 2. Happiness Rings: Weather, Score, and Happiness (Radial Visualization)



sketch2

This sketch, titled “**Happiness Rings: Weather, Score, and Happiness (Radial Visualization)**”, uses **concentric rings** to represent different happiness levels. Data points (circles) are placed on rings corresponding to happiness values, colored by weather.

- **Yellow = Sunny**
- **Gray = Cloudy**
- **Blue = Rainy**
- Rings labeled for happiness levels 7–10
- Golf scores (87–95) written inside circles

Strengths of the Sketch

- The concentric circles create a clean comparison of **absolute happiness scores**.
- Weather color coding is clear and readable.
- A circular layout matches the golf theme visually.
- Score labels add a secondary variable.

Weaknesses

- Uses **happiness level only**, but cannot show “before vs. after” differences.

- Adding both before and after would clutter the rings.
- The focus shifts toward **performance score**, which is *not* the project's main research question.
- Hard to visualize emotional *change*—only static values.

Why I Did Not Use This Sketch

Although attractive and well-structured, the sketch does **not emphasize the core story** of the project: happiness *change*. The final radial “spoke” design makes before/after contrast extremely clear, which directly serves my hypothesis.

For this reason, I selected the spoke-based radial visualization as the final form.

6. Rationale for Final Visualization

6.1 Why This Format Works Best

Based on readings from **Data-to-Viz**, class lectures, and course examples, the radial design meets all goals of the project:

- Allows **comparison** of before vs. after values
- Encodes **weather** clearly and consistently
- Handles **23 observations** without overlap
- Communicates emotional “movement” outward (before → after)
- Fits easily within the 1600×1000 requirement

6.2 Color & Shape Choices

- **Blue = Rainy** → cool tone
- **Orange = Sunny** → warm, energetic
- **Gray = Cloudy** → neutral, muted
- **Thin line segments** → reduce clutter
- **Small circular markers** → clear endpoints
- **Golf ball in center** → reinforces theme visually

6.3 Why Other Chart Types Were Not Selected

- **Timeline** → emphasizes date, not happiness change
 - **Bar chart** → rigid, loses circular golf metaphor
 - **Scatterplot** → cannot show two happiness values per round cleanly
 - **Heatmap** → poor for small datasets
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7. Technical Implementation (JavaScript & D3)

Your project requires documenting your code.

Below is the complete explanation, in the format required (Summary / Requirements / Parameters / Returns).

7.1 Function: loadData()

Summary: Loads data.json using `d3.json()` and sends it to the visualization.

Requirements: data.json must be in the same directory.

Returns: None.

Notes: Required by course policy (must not use fetch).

7.2 Function: classifyWeather(weatherString)

Summary: Converts raw text descriptions into one of three categories.

Requirements: weatherString must be a non-empty string.

Parameters:

- weatherString (String): description from dataset
Returns: "Sunny", "Cloudy", or "Rainy"
-

7.3 Function: buildVisualization(rawData)

Summary: Main function that builds the entire radial diagram.

Requirements: Must be called after JSON loads.

Parameters:

- rawData (Array): 23 objects containing date, course, weather, before, after
Returns: None.

Main actions:

- Calculate angle step
- Convert weather to categories
- Create scales for happiness → radius
- Draw before/after segments
- Draw weather markers
- Draw center golf ball
- Draw legend
- Style axes and layout

8. Findings & Conclusion

8.1 Key Observations

Based on the visualization:

- **Sunny / partly sunny days produced the largest happiness increases**, often 3–4 points.
- **Cloudy days produced moderate improvements.**
- **Rainy days were inconsistent**—some rounds showed strong improvement, some small.
- In all 23 rounds, **happiness never decreased** after golf.

8.2 Interpretation

My hypothesis is **mostly supported**:

- Sunny → strong positive emotional change
- Cloudy → smaller but noticeable change
- Rainy → mixed

8.3 Surprising Finding

I expected rainy days to always be low happiness, but several rainy rounds had large improvements (e.g., from 5 to 10). This indicates golf itself has a strong positive emotional impact regardless of weather.

9. Limitations & Future Data

Limitations

- Dataset is small (23 rounds).
- Weather classifications are manually simplified.
- No measurement of sleep, stress, golf score, or physical condition.
- Only rounds from Week 1–12 are included.

Future Data to Collect

- Score / performance
- Wind speed, temperature
- Walking vs. cart
- Time of day

- Social factors (playing alone vs. with partner)

This would allow deeper causal analysis.

10. Self-Analysis

This project changed the way I think about data, emotion, and personal habits. Through the quarter, I learned:

- Visualization reveals patterns I could not see before
- Weather affects me strongly, but golf boosts happiness regardless
- Collecting data regularly increases self-awareness
- Simple data, when visualized well, becomes meaningful

Before this class, I rarely tracked mood or behavior. Now, I am motivated to continue collecting personal data as a tool for self-reflection and behavioral improvement.

11. Works Cited (MLA Format)

Holtz, Yan. *Data to Viz*. www.data-to-viz.com.
Taylor-Laird, Jay. *DGM 6109 Course Slides, Weeks 7–12*. Northeastern University, 2025.
Healy, Kieran. *Data Visualization: A Practical Introduction*. Princeton University Press, 2018.
Few, Stephen. *Information Dashboard Design*. Analytics Press, 2013.

Appendix A — Full Dataset (23 Rounds)

(All entries from *data.json*)

Date	Course	Weather	Happiness Before	Happiness After
2025-09-14	University Golf Club	Overcast (15 °C)	6	8
2025-09-17	Mayfair Golf Course	Cloudy (14 °C)	6	8

2025-09-21	Northview Golf & Country Club	Partly Sunny (16 °C)	7	10
2025-09-26	Quilchena Golf Course	Light Rain (13 °C)	6	9
2025-09-29	University Golf Club	Mostly Cloudy (12 °C)	5	7
2025-10-03	Mayfair Golf Course	Drizzle, Cool (12 °C)	5	8
2025-10-08	Northview Golf & Country Club	Cloudy (13 °C)	6	8
2025-10-11	Quilchena Golf Course	Partly Sunny (15 °C)	7	10
2025-10-29	Quilchena Golf Course	Cloudy (15 °C)	6	9
2025-11-02	Langara Golf Course	Partly Sunny (18 °C)	4	8
2025-11-05	Quilchena Golf Course	Sunny (15 °C)	6	7
2025-11-07	McCleery Golf Course	Partly Sunny (17 °C)	7	8
2025-11-09	Northlands Golf Course	Light Rain (13 °C)	5	10
2025-11-12	Quilchena Golf Course	Light Rain (15 °C)	6	9
2025-11-15	Quilchena Golf Course	Partly Sunny (13 °C)	4	10
2025-11-16	Mayfair Golf Course	Rainy (13 °C)	6	8
2025-11-18	Quilchena Golf Course	Cloudy (16 °C)	5	8
2025-11-23	Myfair Golf Course	Rainy (10 °C)	6	9
2025-12-01	Quilchena Golf Course	Light Rain (12 °C)	5	7
2025-12-03	Quilchena Golf Course	Cloudy (11 °C)	6	8

2025-12-05	Quilchena Golf Course	Light Rain (13 °C)	5	7
2025-12-07	Kings Link Golf Course	Light Rain (13 °C)	4	9